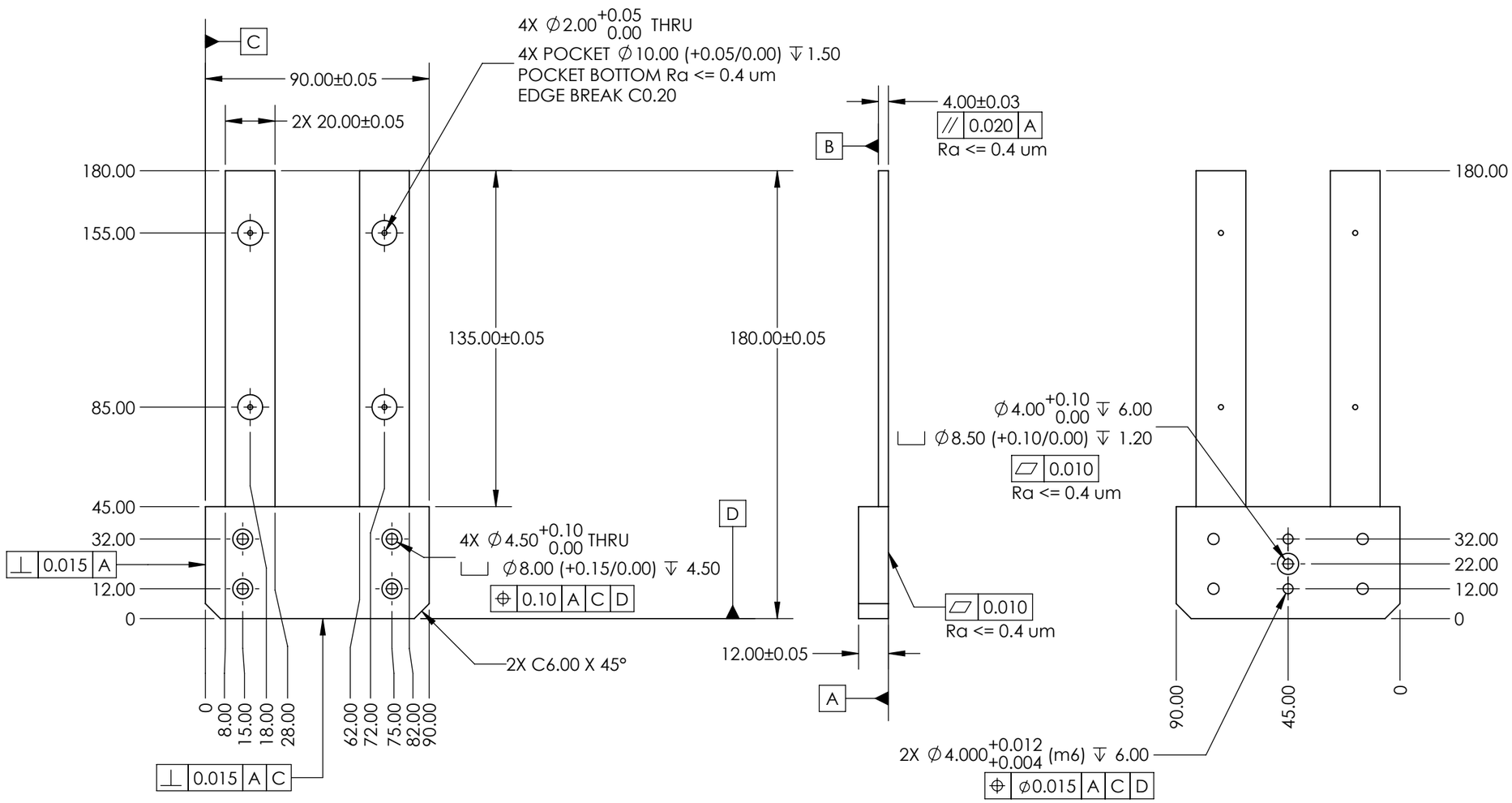
GENERAL NOTES: UNLESS OTHERWISE SPECIFIED

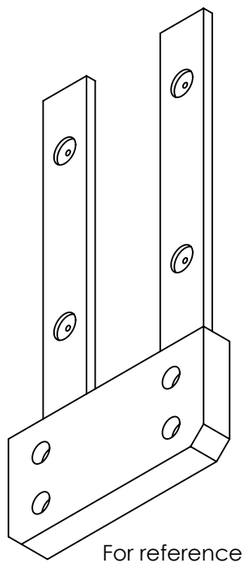
1. GENERAL & STANDARDS COMPLIANCE:
1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
1.2 3D CAD DATASET (W3-SEMI-WAFER-EFFECTOR-6.3.STEP) GOVERNS NOMINAL GEOMETRY FOR UNTOLERANCED SURFACES.
1.3 GENERAL TOLERANCES: LINEAR +/-0.05 mm, ANGULAR +/-0.2 DEG (HIGH-SPEED 5-AXIS CNC MILLING).
1.4 THIRD ANGLE PROJECTION APPLIES THROUGHOUT. ALL UNITS IN MILLIMETERS.
1.5 COORDINATE REFERENCE FRAME: FIRST OCTANT CONVENTION (ALL POSITIVE X, Y, Z FROM DATUM ORIGIN C-D-A).

2. CLEANROOM (CLASS 1 / ISO 1) & ZERO-CONTAMINATION ASSURANCE:
2.1 MATERIAL OPTION A: CARBON-FIBER REINFORCED PEEK (ESD PEEK-CA30), SURFACE RESISTIVITY 10^6 - 10^9 OHM/SQ TO PREVENT ELECTROSTATIC DISCHARGE (ESD) WAFER DAMAGE.
2.2 MATERIAL OPTION B: FORGED AEROSPACE ALUMINUM 7075-T6 (AMS 4045).
2.3 100% SUBTRACTIVE 5-AXIS CNC MACHINING ONLY. STRICTLY PROHIBIT ADDITIVE MANUFACTURING (SLM/SLS) OR SINTERING TO PREVENT PARTICLE SHEDDING IN VACUUM CLUSTER TOOLS.

3. WAFER BACKSIDE ZERO-MARKING & DYNAMIC CONTROLS:
3.1 PRIMARY DATUM [A] (ROBOT FLANGE MOUNTING FACE): FLATNESS <= 0.010 mm, SURFACE FINISH Ra <= 0.4 MICRONS.
3.2 SECONDARY DATUM [B] (WAFER SEATING SURFACES): PARALLEL TO [A] WITHIN 0.020 mm, Ra <= 0.4 MICRONS TO ELIMINATE LOCAL DIE STRESS ON THIN WAFERS (< 100 µm).
3.3 DYNAMIC STIFFNESS & DEFLECTION: UNDER 300mm WAFER GRAVITY + 3.0G TRANSFER ACCELERATION, CANTILEVER TIP SAG SHALL NOT EXCEED 0.15 mm.
3.4 4X SUCTION POCKETS (Ø10.00 mm x 1.50 mm DEEP): EDGES CHAMFERED C0.20 mm UNDER 20X MAGNIFICATION TO ELIMINATE BURRS AND CONTACT SCRATCHES.

4. CLEANING & SURFACE FINISH:
4.1 SURFACE FINISH FOR AL 7075: HARD ANODIZE TYPE III (MIL-A-8625, 20-25 MICRONS THICK) WITH PTFE IMPREGNATION FOR ULTRA-LOW OUTGASSING.
4.2 FINAL CLEANING: MULTI-STAGE ULTRASONIC DEGREASING USING DEIONIZED WATER (18 MOhm·cm), CLEANROOM PACKAGED WITH DOUBLE PE BAGS (ZERO PARTICLES > 0.3 µm PERMITTED).




For reference

UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: Linear ±0.05 mm, Angular ±0.2° (High-Speed 5-Axis CNC Milling)		FINISH: For Al 7075: Hard Anodize Type III (MIL-A-8625, 20-25 µm, Ultra-Clean PTFE Sealed) / For PEEK: Precision Ultrasonic Degreased Class 1 Cleanroom Packaged		DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING	REVISION: A
	NAME	SIGNATURE	DATE			TITLE: 300mm WAFER HIGH-SPEED TRANSFER END-EFFECTOR VACUUM CHUCK	
DRAWN							
CHK'D							
APPV'D							
MFG							
Q.A				MATERIAL: Option A: ESD PEEK-CA30 (Carbon-Fiber Reinforced, Surface Resistivity 10^6 - 10^9 Ω/sq) / Option B: Forged Aerospace Aluminum 7075-T6 (AMS 4045)		DWG NO. W3-SEMI-WAFER-EFFECTOR-6.3	A3
				WEIGHT: ~94.6 g (PEEK-CA30) / ~190.0 g (Al 7075-T6)		SCALE NTS	SHEET 1 OF 1